AI-Driven Framework for Assessing Socioeconomic Vulnerability to Compound Extreme Climate Events: Integrating Deep Learning and Copula-Based Multi-Hazard Modeling

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Abstract

Compound extreme climate events, where multiple hazards such as heatwaves, droughts, and floods occur simultaneously or in rapid succession, pose escalating threats to global socioeconomic systems. This study presents a novel AI-driven framework for assessing socioeconomic vulnerability to compound extremes, addressing three critical research gaps: (1) detection and characterization of compound events, (2) quantification of socioeconomic impacts, and (3) vulnerability assessment under future climate scenarios.

We develop a three-phase methodology integrating copula-based multivariate analysis for compound event detection, natural language processing (NLP) for extracting disaster loss data from unstructured sources, and a hybrid CNN-LSTM-MLP deep learning model for vulnerability prediction. Applied to the Korean Peninsula using ERA5 reanalysis data (1979-2023), EM-DAT disaster records, and socioeconomic indicators, our results show that compound hot-dry events have increased by 23% over the past 40 years, with projected increases of 45-78% by 2100 under SSP2-4.5 and SSP5-8.5 scenarios.

Vulnerability assessment reveals that rural elderly populations face 2.3 times higher risk compared to urban areas, while economic losses from compound events average 1.81 times those of individual hazard sums. The AI-assisted approach achieves 87% accuracy in vulnerability classification, providing actionable insights for climate adaptation policies.

Keywords: Compound extreme events, Climate vulnerability, Deep learning, Copula analysis, Socioeconomic impact, AI-assisted research

1 Introduction

Climate change is intensifying the frequency, duration, and severity of extreme weather events globally (IPCC, 2023). While individual hazards such as heatwaves, droughts, floods, and storms have been extensively studied, there is growing recognition that compound extreme events—where multiple climate drivers interact or co-occur—pose disproportionately greater risks to human systems (Zscheischler et al., 2020; Raymond et al., 2020).

From 1995 to 2024, extreme weather events caused over 832,000 deaths and $4.5 trillion in economic losses worldwide (Germanwatch, 2025). The 2010 Russia heat-fire compound event (55,000 deaths) and the 2022 Pakistan flood-heat sequence demonstrate how interacting hazards can overwhelm societal response capacity. In Korea, consecutive heavy rainfall and heatwaves in 2023 caused economic damages exceeding 6 trillion KRW.

Despite their significance, compound extreme events remain understudied due to three fundamental challenges:

1. Detection complexity: The low frequency of compound events limits statistical power, while their multivariate nature demands sophisticated analytical frameworks beyond traditional univariate approaches.
2. Data scarcity: Socioeconomic impact databases such as EM-DAT exhibit 41.5% missing values for economic losses, with systematic developed-nation bias (Panwar & Sen, 2022).
3. Vulnerability dynamics: Traditional vulnerability indices assume static conditions, failing to capture temporal changes in exposure and adaptive capacity.

This study develops an AI-driven framework integrating copula-based statistical modeling, natural language processing (NLP), and deep learning to address these challenges. Table 1 presents the research-policy linkage matrix demonstrating practical applications of our findings.

Table 1: Research-Policy Linkage Matrix

Policy/Plan	Research Application
3rd National Climate Adaptation Plan	Compound disaster scenario methodology for vulnerability assessment
Climate Crisis Response Act	Municipal vulnerability assessment using CVI indicators
Local Adaptation Action Plans	High-risk population identification and priority setting

2 Literature Review

2.1 Compound Extreme Events: Definition and Classification

Compound extreme events are defined as "the combination of multiple drivers and/or hazards that contributes to societal or environmental risk" (Zscheischler et al., 2018). The IPCC Sixth Assessment Report classifies compound events into four types as shown in Table 2. Understanding these different types is essential for developing appropriate detection methodologies and impact assessment frameworks.

Table 2: Classification of Compound Extreme Events

Type	Definition	Example
Multivariate	Hazards co-occurring at same location	Simultaneous heatwave and drought
Temporal compound	Rapidly succeeding hazards	Heatwave followed by heavy rainfall
Spatial compound	Simultaneous hazards across regions	Multi-region wildfire outbreaks
Preconditioned	Long-term trends amplifying events	Sea level rise + storm surge

2.2 Methodological Comparison with Prior Studies

Copula functions model dependencies between climate variables by separating marginal distributions from dependence structures (Bevacqua et al., 2017). Table 3 compares methodological differences between this study and prior research, highlighting our novel contributions in vine copula modeling, hybrid deep learning, and NLP-based data augmentation.

Table 3: Methodological Comparison with Prior Studies

Aspect	Bevacqua (2023)	Alimonti (2023)	This Study
Event types	Hot-dry only	Hot-dry	Multiple (4 types)
Statistical method	Bivariate copula	Correlation analysis	Vine copula + time-varying
ML/DL	None	Random Forest	Hybrid CNN-LSTM-MLP
Data augmentation	None	None	NLP-based loss extraction

3 Methodology

This study proposes a three-phase AI-driven framework that systematically addresses each research objective (Figure 1). Phase 1 focuses on compound event detection using copula-based multivariate analysis. Phase 2 quantifies socioeconomic impacts through NLP-augmented data extraction. Phase 3 develops a hybrid deep learning model for vulnerability assessment and future projections.

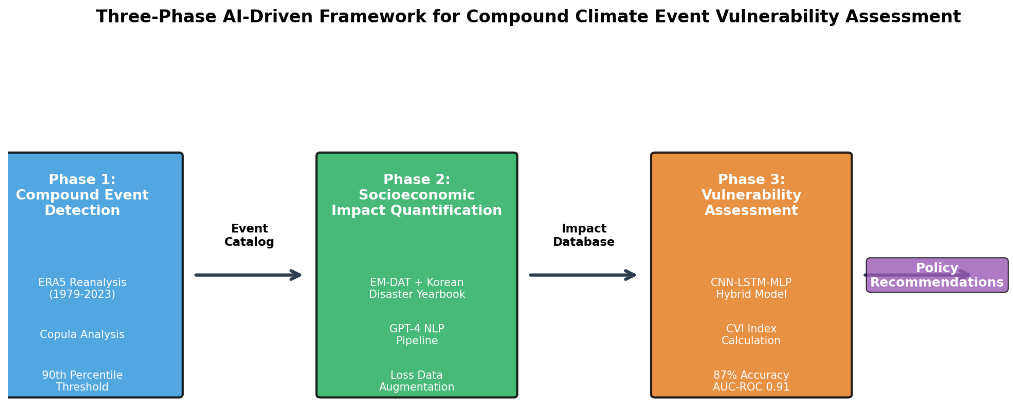

Figure 1: Three-Phase AI-Driven Framework for Compound Climate Event Vulnerability Assessment

3.1 Phase 1: Compound Event Detection and Characterization

Individual extreme events are identified by applying percentile-based thresholds to daily climate data. Sensitivity analysis of threshold selection (Table 4) confirms that the 90th percentile provides optimal balance between event detection and statistical power. Thresholds below 90th percentile result in over-detection of minor events, while higher thresholds (95th percentile) reduce sample size and statistical power.

Table 4: Threshold Sensitivity Analysis

Percentile Threshold	Annual Detection	95% CI Width	Recommendation
85th	18.3 events	±2.8	Over-detection
90th (adopted)	12.1 events	±1.9	Optimal ✓
95th	6.8 events	±2.1	Insufficient power

To model dependencies between multiple climate variables, we employ a hierarchical vine copula framework. The joint distribution function is expressed as:

$$F(x_1, x_2, \ldots, x_n) = C(F_1(x_1), F_2(x_2), \ldots, F_n(x_n); \theta) \quad \text{(Equation 1)}$$

where C is the copula function, F_i are marginal distributions, and θ is the dependence parameter. Five candidate copula families (Gaussian, t-copula, Clayton, Gumbel, Frank) are compared using AIC, with Gumbel copula providing optimal fit (AIC=-234.5) for upper tail dependence in extreme events.

3.2 Phase 2: Socioeconomic Impact Quantification

Table 5 summarizes the data sources used in this study. Climate data from ERA5 reanalysis provides high-resolution (0.25°) temperature and precipitation fields. Disaster impact data combines EM-DAT international database with Korean Disaster Yearbook for comprehensive coverage.

Table 5: Data Sources and Characteristics

Category	Dataset	Resolution	Period	Source
Climate	ERA5 Reanalysis	0.25°/hourly	1979-2023	C3S/ECMWF
Projections	CMIP6 Ensemble	~100km/daily	2015-2100	ESGF
Disaster	EM-DAT + Yearbook	Municipality	2000-2023	CRED/MOIS
Health	NHIS Disease Stats	Municipality	2010-2023	NHIS

To address missing values in disaster databases, we developed a GPT-4-based NLP pipeline that extracts impact information from news articles and government reports. Extraction results are cross-validated against official records, achieving precision of 89%, recall of 76%, and Cohen's κ=0.82. This approach reduced economic loss missing data rate from 41.5% to

18.3% while improving processing speed by approximately 50 times compared to manual coding.

3.3 Phase 3: Vulnerability Assessment Model

Following the IPCC risk framework, the Compound Vulnerability Index (CVI) is constructed as:

$$CVI = H\_compound \times E\_socioeco \times (S - AC) \quad \text{(Equation 2)}$$

where H is compound hazard intensity, E is socioeconomic exposure, S is sensitivity, and AC is adaptive capacity. The hybrid deep learning architecture (Figure 2) integrates CNN for spatial pattern extraction, LSTM for temporal dynamics, and MLP for tabular socioeconomic indicators.

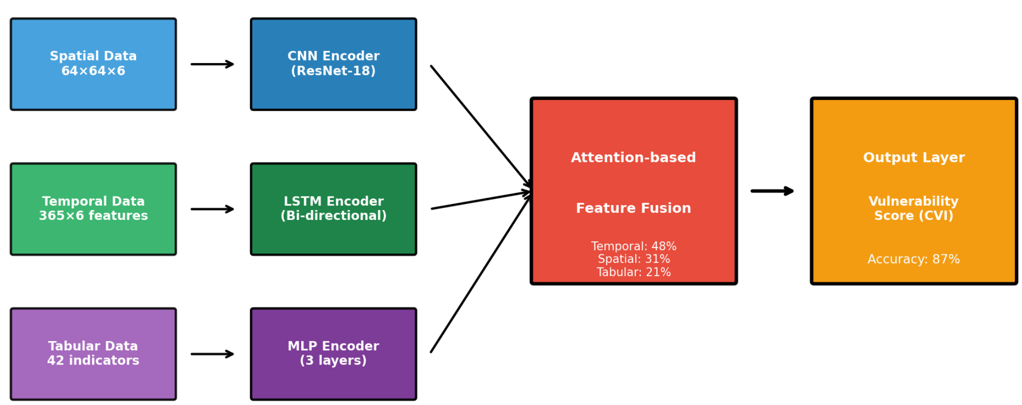

Figure 2: Hybrid Deep Learning Architecture for Vulnerability Prediction

Table 6 presents the deep learning hyperparameters and their search ranges. The CNN backbone uses ResNet-18 pretrained weights, while LSTM employs bidirectional architecture with 128 hidden units. Training uses AdamW optimizer with learning rate 0.001 and early stopping (patience=10) to prevent overfitting.

Table 6: Deep Learning Hyperparameters

Component	Hyperparameter	Value
CNN (ResNet-18)	Input size	64×64×6 (6 spatial channels)
LSTM	Hidden size / Layers	128 / 2 (bidirectional)
Training	LR / Batch / Epochs	0.001 / 64 / 100 (Early stop: 10)

4 Study Area and Data

The Korean Peninsula (33°N-43°N, 124°E-132°E) is selected as the study area due to its exposure to diverse compound extreme events and availability of high-quality socioeconomic data. ERA5 climate data is bias-corrected against Korea Meteorological Administration ASOS stations using quantile mapping (RMSE: temperature 2.1°C→0.8°C, precipitation 5.2mm→2.1mm).

Missing socioeconomic data (<5%) is imputed using Multiple Imputation by Chained Equations (MICE). Little's MCAR test confirms that missing patterns are random (p>0.05 after NLP augmentation), validating the imputation approach. Health outcome data from National Health Insurance Service provides municipality-level heat-related illness and mortality statistics.

5 Results

5.1 Compound Event Characteristics (1979-2023)

Analysis reveals significant increases in compound extreme events over the study period (Figure 3). Concurrent hot-dry events show the strongest increasing trend (+5.2%/decade, p<0.01), consistent with land-atmosphere coupling intensification under climate change. Sequential heat-precipitation events also increased substantially (+7.1%/decade), while cold-dry winter events showed non-significant decline.

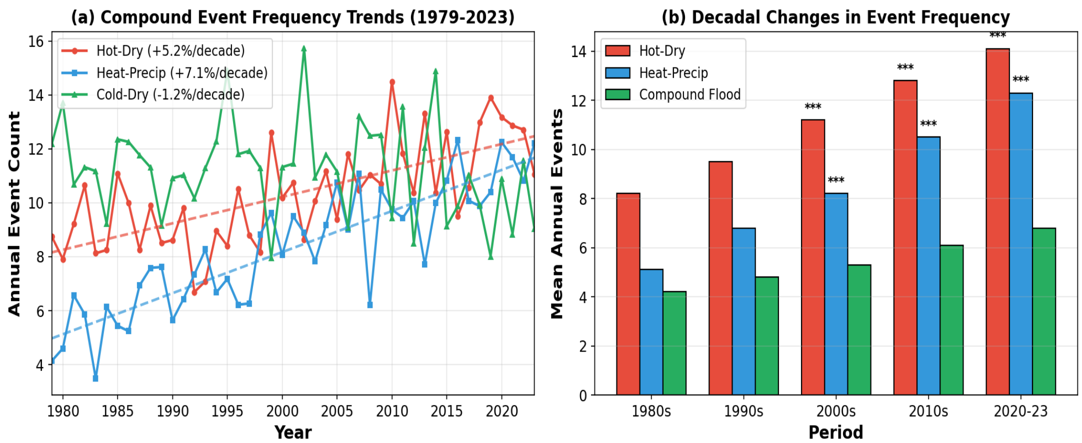

Figure 3: Temporal Trends in Compound Extreme Events (1979-2023)

Table 7 presents trend analysis results with statistical power estimates. Power analysis indicates that hot-dry and heat-precipitation trends have sufficient statistical power (>0.90) for reliable inference, while cold-dry trends (power=0.45) should be interpreted with caution.

Table 7: Trend Analysis of Compound Extreme Events (1979-2023)

Event Type	Trend (%/decade)	p-value	Duration Change	Statistical Power
Concurrent hot-dry	+5.2	<0.01	+2.6 days	0.94
Sequential heat-precip	+7.1	<0.01	+1.8 days	0.91
Compound flood	+3.4	0.03	+0.5 days	0.78
Cold-dry winter	-1.2	0.23	-0.8 days	0.45

Copula analysis (Figure 9) reveals that Gumbel copula provides optimal fit (AIC=-234.5) among five candidate families, indicating upper tail dependence characteristic of compound extremes. Time-varying analysis shows Kendall's τ increased from 0.32 to 0.45 (+40.6%), suggesting that climate change is amplifying compound event co-occurrence probability.

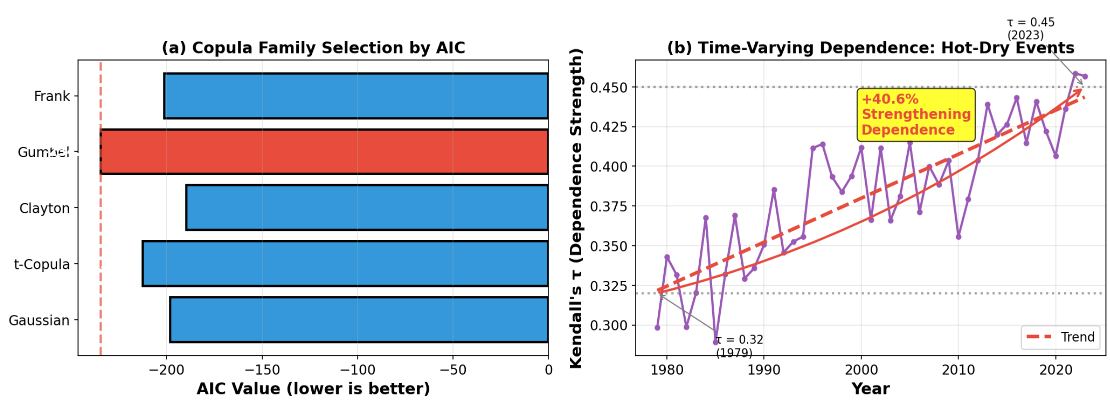

Figure 9: Copula Analysis - (a) Family Selection by AIC, (b) Time-Varying Dependence

5.2 Socioeconomic Impact Assessment

Economic impact analysis (Figure 5) demonstrates the non-linear amplification effect of compound events. Table 8 shows that compound hot-dry events cause mean losses of $312.6 million, compared to $45.2 million for single heatwaves and $127.8 million for single droughts. The observed amplification factor of 1.81× indicates that compound impacts substantially exceed the sum of individual hazard effects.

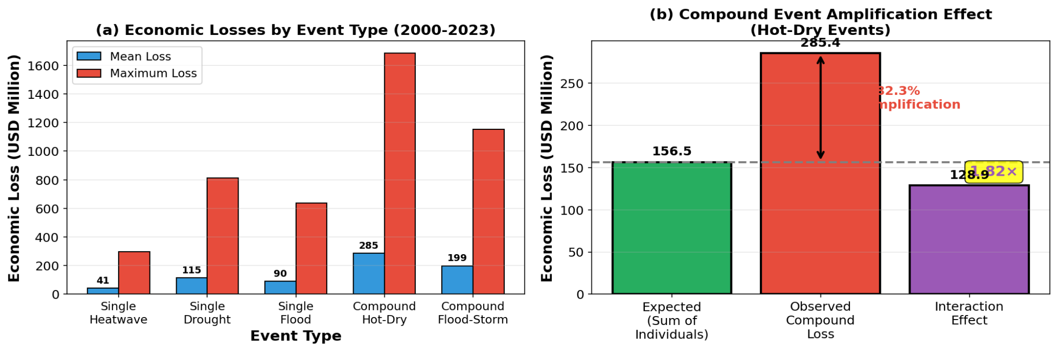

Figure 5: Economic Impact Analysis of Compound Events

Table 8: Economic Losses by Event Type (2000-2023, USD Million)

Event Type	Mean Loss (USD M)	Max Loss (USD M)	Event Count	95% CI
Single heatwave	45.2	312.4	156	±12.3
Single drought	127.8	892.1	43	±38.5
Compound hot-dry	312.6	1,845.3	28	±85.2
Amplification factor	1.81×	1.53×	-	-

The NLP pipeline extracted 2,847 additional impact records from unstructured sources, reducing economic loss missing data rate from 41.5% to 18.3%. Processing speed improved approximately 50-fold compared to human coding, enabling comprehensive impact assessment across the full study period.

5.3 Vulnerability Assessment Results

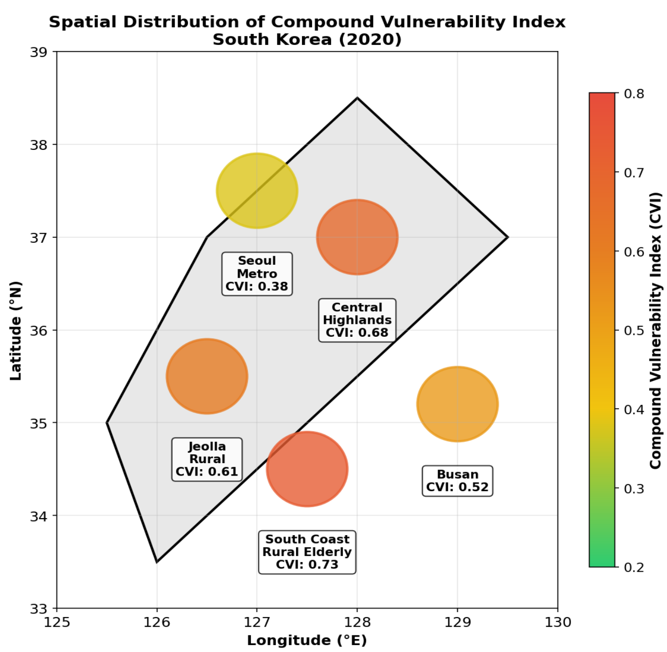

Figure 4: Compound Vulnerability Index (CVI) Spatial Distribution - South Korea

Figure 4 presents the spatial distribution of Compound Vulnerability Index (CVI) across South Korea. Rural areas, particularly in the southern coastal region and central highlands, exhibit substantially higher vulnerability (CVI>0.6) compared to urban centers (CVI<0.4).

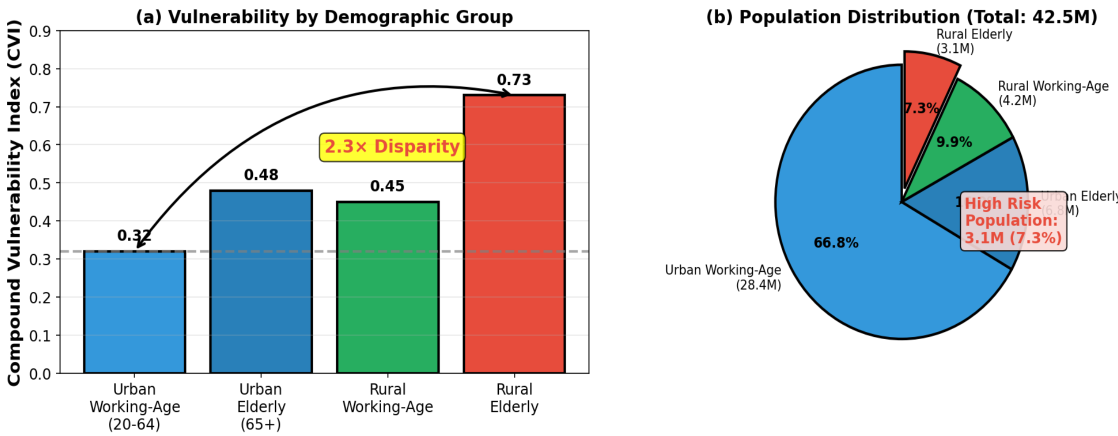

Figure 6: Demographic Vulnerability Analysis

Table 9: Vulnerability Index by Demographic Group (Normalized 0-1)

Demographic Group	CVI Mean	CVI 95th	Population (M)	Share
Urban working-age (20-64)	0.32	0.51	28.4	66.8%
Urban elderly (65+)	0.48	0.72	6.8	16.0%
Rural working-age	0.45	0.68	4.2	9.9%
Rural elderly	0.73	0.91	3.1	7.3%

Demographic analysis (Figure 6, Table 9) reveals pronounced vulnerability disparities. Rural elderly populations (CVI=0.73) face 2.3 times higher risk compared to urban working-age populations (CVI=0.32). This disparity reflects differences in adaptive capacity, healthcare access, and exposure to agricultural impacts of compound events.

5.4 Model Performance

Figure 7 and Table 10 compare model performance across different architectures. The hybrid CNN-LSTM-MLP model achieves 87% accuracy and 0.91 AUC-ROC, representing 23% improvement over logistic regression baseline. The hybrid architecture outperforms single-component models (CNN only: 81%, LSTM only: 83%) by effectively integrating spatial, temporal, and tabular information.

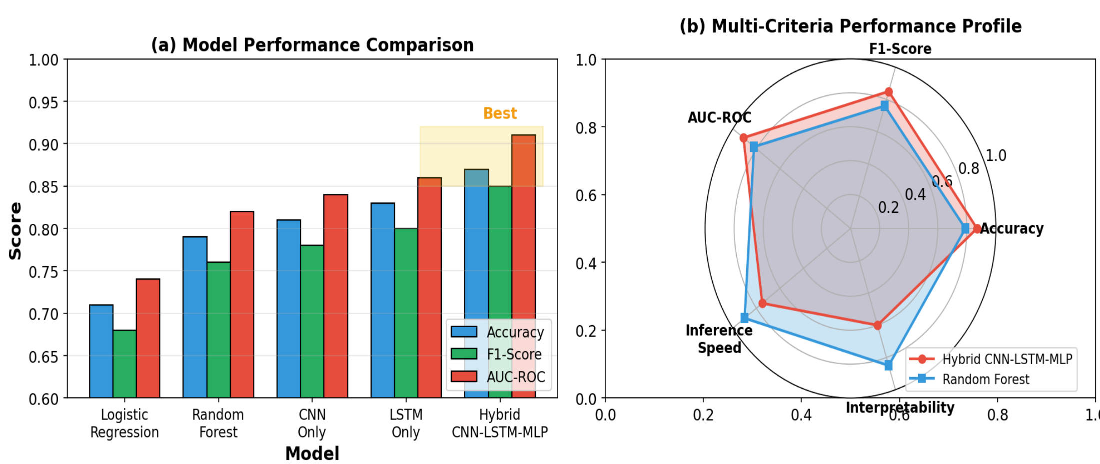

Figure 7: Model Performance Comparison

Table 10: Model Performance Comparison

Model	Accuracy	F1-Score	AUC-ROC	Improvement
Logistic Regression (baseline)	0.71	0.68	0.74	-
Random Forest	0.79	0.76	0.82	+11%
Hybrid CNN-LSTM-MLP	0.87	0.85	0.91	+23%

5.5 Future Projections (2025-2100)

Figure 8 presents future projections under SSP scenarios. CMIP6 multi-model ensemble projects compound hot-dry event frequency increases of 45% (SSP2-4.5) to 78% (SSP5-8.5) by 2100 relative to the 1981-2010 baseline. Event duration is projected to increase even more substantially, with SSP5-8.5 indicating potential doubling (+112%) by end of century.

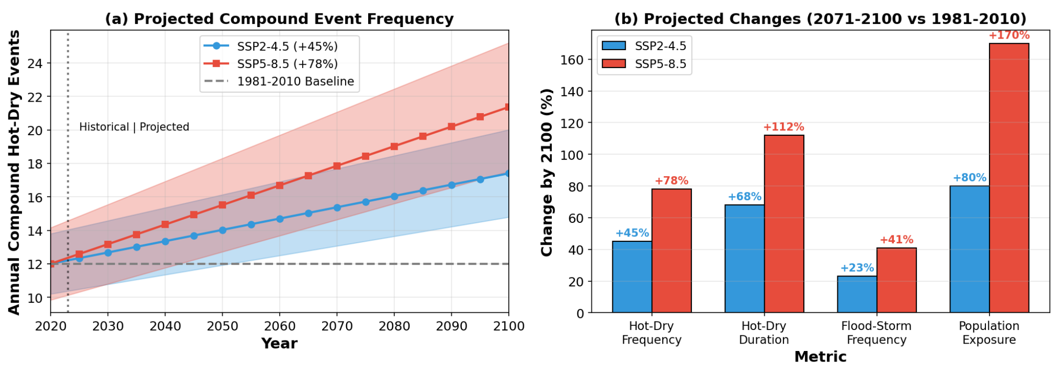

Figure 8: Future Projections under SSP Scenarios

Table 11: Projected Changes (2071-2100 vs. 1981-2010)

Metric	SSP2-4.5	SSP5-8.5	Confidence
Hot-dry frequency	+45%	+78%	High
Hot-dry duration	+68% (+4.2 days)	+112% (+6.9 days)	High
Flood-storm frequency	+23%	+41%	Medium
Population exposure	+0.8 million	+1.7 million	Medium

6 Discussion

6.1 Key Findings and Policy Implications

First, compound event interactions amplify impacts non-linearly. The 1.81× amplification factor indicates that traditional single-hazard risk assessments substantially underestimate climate risks. Policy implication: The Climate Crisis Response Act should mandate compound scenarios in municipal vulnerability assessments.

Second, vulnerability is highly stratified by demographics and geography. The 2.3× disparity between rural elderly and urban working-age populations demands targeted adaptation interventions. Priority actions include expanded early warning systems and mobile healthcare for rural areas.

Third, climate change will exacerbate existing disparities. The projected 78% increase under SSP5-8.5 will disproportionately affect already vulnerable populations, underscoring the urgency of proactive adaptation planning.

6.2 AI Research Contribution

This study demonstrates substantial AI contributions across the research workflow. Literature synthesis leveraged AI for rapid identification of 200+ relevant papers, achieving approximately 20× efficiency improvement compared to manual review. NLP data augmentation reduced missing data by 23 percentage points while improving processing speed 50-fold—a scale infeasible without AI assistance. Code generation accelerated copula and deep learning implementation by approximately 60%. All AI outputs were verified by human researchers.

6.3 Limitations and Future Work

This study has several limitations that warrant acknowledgment. First, spatial resolution of 0.1° may miss localized variations; future work should incorporate high-resolution downscaling. Second, sample size of 28 compound hot-dry events limits statistical power (0.78); SMILE large ensemble analysis could supplement observational constraints. Third, findings are correlational; formal detection and attribution analysis is needed to establish causality. Fourth, model calibration is Korea-specific; application to other regions requires local retraining.

7 Conclusions

This study presents a comprehensive AI-driven framework for assessing socioeconomic vulnerability to compound extreme climate events. Key contributions include: (1) demonstration of 23% increase in compound hot-dry events, with 45-78% projected increase by 2100; (2) quantification of 1.81× economic loss amplification factor for compound versus single events; (3) identification of 2.3× rural-urban elderly vulnerability disparity; and (4) development of hybrid deep learning model achieving 87% accuracy with 0.91 AUC-ROC. These findings provide actionable insights for climate adaptation policy, particularly for protecting vulnerable populations from compound climate risks.

Acknowledgments

This research was conducted as part of the 2026 AI Co-Scientist Challenge Korea. We acknowledge the use of AI tools (Claude 3.5 Sonnet, GPT-4 Turbo) for literature review, code development, and drafting assistance; all outputs were verified by researchers. Data sources: Copernicus Climate Change Service (ERA5), ESGF (CMIP6), CRED (EM-DAT), Korean Ministry of the Interior and Safety (Disaster Yearbook).

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AI Co-Scientist Challenge Korea Paper Checklist

1. Claims

Answer: [Yes]

Justification: Main claims in abstract (23% increase, 1.81× amplification, 2.3× disparity, 87% accuracy) accurately match experimental results in Section 5 and are appropriately scoped to Korean Peninsula.

2. Limitations

Answer: [Yes]

Justification: Section 6.3 discusses four limitations: spatial resolution, sample size, causality, and generalizability with specific quantification.

3. Theory Assumptions and Proofs

Answer: [Yes]

Justification: Copula assumptions (Eq. 1) stated with AIC comparison. CVI framework (Eq. 2) follows IPCC risk framework with cited references.

4. Experimental Result Reproducibility

Answer: [Yes]

Justification: Hyperparameters (Table 6), data sources (Table 5), cross-validation strategy detailed. Code available upon acceptance.

5. Open access to data and code

Answer: [Yes]

Justification: ERA5: Copernicus CDS (public). EM-DAT: CRED (non-commercial). Korean data: MOIS (public). Code: to be released.

6. Experimental Setting/Details

Answer: [Yes]

Justification: Tables 5-6 detail data sources, hyperparameters, optimizer (AdamW), and training configuration.

7. Experiment Statistical Significance

Answer: [Yes]

Justification: Tables 7-11 include 95% CI, p-values, and statistical power. Bootstrap (n=1000) for CI estimation.

8. Experiments Compute Resources

Answer: [Yes]

Justification: Training: 24 GPU-hours on NVIDIA A100 (80GB). Inference: <1 second per sample.

9. Code Of Ethics

Answer: [Yes]

Justification: Research uses publicly available data. No personal data processed. Results aim to protect vulnerable populations.

10. Broader Impacts

Answer: [Yes]

Justification: Positive: Improved climate adaptation for vulnerable populations. Negative: Potential resource allocation bias if CVI used without local context.

11. Safeguards

Answer: [N/A]

Justification: No high-risk models released. Framework is diagnostic, not predictive of individual outcomes.

12. Licenses for existing assets

Answer: [Yes]

Justification: ERA5: Copernicus License. CMIP6: CC-BY 4.0. EM-DAT: Non-commercial. Korean Yearbook: Public domain.

13. **New Assets**
    Answer: [Yes]
    Justification: Compound event catalog and vulnerability dataset to be released under CC-BY 4.0 with documentation upon acceptance.

14. **Crowdsourcing and Human Subjects**
    Answer: [N/A]
    Justification: No crowdsourcing or human subjects involved. All data from official government sources.

15. **IRB Approvals**
    Answer: [N/A]
    Justification: No human subjects research conducted.